Tairan Wang

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Education

University College London London, UK
MEng Mathematical Computation, **First-Class Honours (78%)** Sep 2022 – Jun 2026

Research Experience

Back to the Future: Regressing Readability Features from LLMs
[PDF] [arXiv] (First Author) UCL Oct 2026

- Developed BTTF, an interpretable code readability model combining linear regression with traditional code metrics, embedding-based semantic organisation, and language-model predictability.
- Evaluated on 1,100 samples covering all six human-rated code-readability datasets in our study, achieving **state-of-the-art (SOTA)** overall performance; further validated robustness through controlled code transformations.

When Specifications Conflict: A Symmetry-Based Framework for Measuring LLM Preferences
[PDF] [arXiv] (First Author) UCL Jul 2026

- Proposed a symmetry-based framework for measuring LLM preferences under conflicting specifications, evaluated on 550 executable conflicts across 11 mathematical function families, with extensions to Boolean algebra, code generation, and clinical rules.

Cognitive Modeling of Code Readability: Working Memory and Semantic Recoverability
[PDF] MEng Dissertation UCL Jun 2026

- Developed CognaScore and PreFixScore to model working-memory load and semantic recoverability in code comprehension; evaluated both on function- and file-level human-rated readability datasets.

Meow-Omni 1: A Multimodal Large Language Model for Feline Ethology
[PDF] [NeurIPS 2026] UCL May 2026

- Developed an open-source quad-modal large language model integrating video, audio, physiological time-series, and textual reasoning for feline intent recognition.

HoarePrompt: Structural Reasoning About Program Correctness in Natural Language
[PDF] [ICSE 2026] UCL & PKU May 2024–Jan 2025

- Proposed a framework for verifying program correctness from natural-language specifications using Hoare triples, strongest postcondition calculus, and k-induction for loops.

Professional Experience

CTO, Alephedge
[Website] [Ordinal] Hong Kong Sep 2026 – Present

- Lead the design and development of Ordinal, an AI agent combining general-purpose capabilities with financial research expertise, integrating tools for data analysis, factor evaluation, and strategy backtesting.

Co-Founder & Core Developer, Alephedge Hong Kong May–Sep 2026

- Built and maintained the company intranet and public website.

Summer Research Internship, UCL London Jun–Sep 2025

- Worked on code readability model.

Technical Skills

- **Programming:** Python, Java, C.
- **AI & Machine Learning:** Machine Learning, Deep Learning, Large Language Models (LLMs), multimodal learning, and AI agent development.
- **Software Engineering:** Full-stack development, AI-assisted development workflows, and maintainable software engineering practices.
- **Quantitative Computing:** Factor research, portfolio construction, risk management.

Academic Achievements

- Ranked first in multiple UCL Computer Science modules, including COMP0010 (Software Engineering, **1/150**), COMP0169 (Machine Learning for Visual Computing, **1/130**), and COMP0024 (Artificial Intelligence and Neural Computing, **1/50**).